WORKED SOLUTION

STEP II 2000, Question 12

Probability and Statistics • Sums of Normal Variables

STEP II 2000, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q12

The key idea

Henry is late exactly when his *total* journey time exceeds the time he has before 0900, so for each route we add the two independent normal journey times to get a single normal variable. Standardising turns "late" into "a standard normal exceeds some number", and the two routes can then be compared without any tables, because Φ is increasing. The rest is the law of total probability and Bayes' theorem.

Solution

Comparing the two routes

We assume that, on each route, the train journey and the onward journey are independent.

King's Cross. Let $X_1 \sim N(55, 25)$ be the train time and $Y_1 \sim N(30, 144)$ the onward time. Since they are independent normal variables, their sum is normal, with means and variances adding:

$$T_1 = X_1 + Y_1 \sim N(85,\ 169), \qquad \text{standard deviation } 13.$$

The train leaves at 0715, so Henry has 105 minutes, and he is late when $T_1 > 105$. With $Z \sim N(0, 1)$,

$$\mathrm{P}(\text{late} \mid \text{King's Cross}) = \mathrm{P}\left(Z > \frac{105 - 85}{13}\right) = 1 - \Phi\left(\tfrac{20}{13}\right) = A.$$

Liverpool Street. Similarly $T_2 = X_2 + Y_2$ with $X_2 \sim N(65, 16)$ and $Y_2 \sim N(25, 9)$ independent, so

$$T_2 \sim N(90,\ 25), \qquad \text{standard deviation } 5.$$

This train leaves at 0720, leaving 100 minutes, so

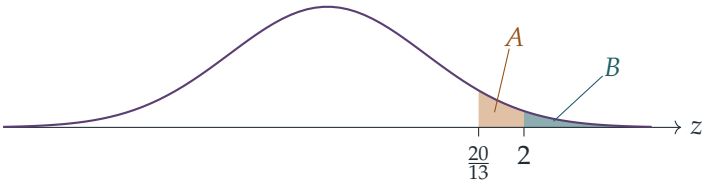
$$\mathrm{P}(\text{late} \mid \text{Liverpool Street}) = \mathrm{P}\left(Z > \frac{100 - 90}{5}\right) = 1 - \Phi(2) = B.$$

Ask yourself How can we compare A and B without tables?

Φ is a cumulative distribution function with a positive density, so it is strictly increasing. Since $\frac{20}{13} < \frac{26}{13} = 2$, we have $\Phi(\frac{20}{13}) < \Phi(2)$, and therefore

$$A = 1 - \Phi\left(\tfrac{20}{13}\right) > 1 - \Phi(2) = B.$$

So Henry is more likely to be late if he catches the King's Cross train, as required. (Interestingly, that route is *faster* on average; it loses because the Underground leg from King's Cross is so variable.)



The shaded tails of the standard normal curve: A is the whole shaded area, B the darker part beyond 2.

The expected number of late arrivals

On a single journey let K be the event that Henry takes the King's Cross train, so $\mathrm{P}(K) = p$, and let D be the event that he is late. By the law of total probability,

$$\mathrm{P}(D) = \mathrm{P}(D \mid K)\,\mathrm{P}(K) + \mathrm{P}(D \mid K')\,\mathrm{P}(K') = pA + (1-p)B.$$

Assuming the M journeys are independent, the number of late arrivals is binomial with parameters M and $\mathrm{P}(D)$, so

$$L = M(pA + (1-p)B) = M(B + p(A-B)).$$

Since $A - B > 0$ and $0 \le p \le 1$, the expression $B + p(A-B)$ increases with p, from B at $p = 0$ to A at $p = 1$. Hence for every possible value of p,

$$BM \le L \le AM,$$

with equality on the left only when $p = 0$ and on the right only when $p = 1$. In words: L/M is a weighted average of A and B, so it must lie between them.

Estimating p

The proportion of late occasions on which he had caught the King's Cross train estimates $\mathrm{P}(K \mid D)$, and because M is large we take the observed $\frac{3}{5}$ as equal to this probability. By Bayes' theorem,

$$\mathrm{P}(K \mid D) = \frac{\mathrm{P}(D \mid K)\,\mathrm{P}(K)}{\mathrm{P}(D)} = \frac{pA}{pA + (1-p)B}.$$

Setting this equal to $\frac{3}{5}$:

$$5pA = 3pA + 3(1-p)B$$
$$2pA + 3pB = 3B,$$

so the estimate is

$$\hat{p} = \frac{3B}{2A + 3B}.$$

Checks

- Numerically, $A \approx 0.0620$ and $B \approx 0.0228$, so King's Cross is late about 6% of the time and Liverpool Street about 2%. A simulation of 4×10^5 journeys on each route agreed to three decimal places.
- With these values $\hat{p} \approx 0.355$. Substituting back, $pA/(pA + (1-p)B) = 0.6$ exactly, and a simulation of the whole process with this p gave about 60% of late arrivals coming from King's Cross.
- Sanity check on the last answer: if the two routes were equally reliable ($A = B$), the late arrivals would reflect only the choice of train, so $\frac{3}{5}$ of them via King's Cross would mean $p = \frac{3}{5}$. The formula gives $3B/5B = \frac{3}{5}$. Because King's Cross is the less reliable route, it is over-represented among late days, so p must be smaller than $\frac{3}{5}$, as it is.

Common slips

- Adding standard deviations instead of variances: the total for King's Cross has variance $25 + 144 = 169$, so its standard deviation is 13, not $5 + 12 = 17$.
- Using the same 105-minute allowance for both trains: the Liverpool Street train leaves five minutes later.
- Setting $p = \frac{3}{5}$ directly. The $\frac{3}{5}$ is $\mathrm{P}(K \mid D)$, not $\mathrm{P}(K)$, and the two differ because the routes are not equally reliable.

Beyond STEP

Two ideas here recur throughout a first probability course. The fact that a sum of independent normal variables is normal is most cleanly proved with moment generating functions, and it is a special case of the stability of the normal family that underlies the central limit theorem. The final part is statistical inference in miniature: we observed an effect (late arrivals) and used Bayes' theorem to infer the hidden cause (the choice of train). Grimmett and Welsh, *Probability: An Introduction*, covers both, and Blitzstein and Hwang, *Introduction to Probability*, has many more Bayes problems of exactly this flavour.

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